

1. Administrative & Study Identification

Full Study Title: A Phase 2 Randomized, Placebo-Controlled, Double-Blind, Parallel-group Study to Evaluate the Efficacy and Safety of MK-1167 as Adjunctive Therapy in Participants With Mild to Moderate Alzheimer's Disease Dementia **Short Title/Acronym:** MK-1167-008 **Protocol Number:** MK-1167-008 **NCT Number:** NCT06721156 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-1167 (with background acetylcholinesterase inhibitor [AChEI] therapy) **Phase of Development:** Phase 2 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if at least one dose level of MK-1167 works to improve cognitive symptoms (memory and thinking) compared to a placebo in participants with mild to moderate Alzheimer's disease (AD) dementia. **Secondary Objectives:** To evaluate the safety, tolerability, and the effect of MK-1167 on the functional ability in activities of daily living (ADCS-ADL) at Weeks 12 and 24.

2.2 Background & Rationale To address the continuing cognitive decline in patients with mild to moderate Alzheimer's disease dementia. AChEI therapy is the standard treatment for AD dementia. This study investigates whether giving MK-1167, an experimental oral medication, alongside standard AChEI therapy can further improve memory, mental activity, and daily functioning.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Quadruple-blind (Participant, Care Provider, Investigator, Outcomes Assessor) **Randomization:** Parallel Assignment (MK-1167 0.3 mg, 1 mg, 3 mg, or Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 100 to 500 participants **Number of Sites:** Multicenter **Estimated Study Start:** 16-Dec-2024 **Estimated Primary Completion:** 31-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 55 to 90 years old. 2. Has mild to moderate Alzheimer's Disease (AD) dementia (i.e., Stage 4 or

Stage 5 AD) based on the Alzheimer's Association Revised Criteria. 3. Has a Mini-Mental State Examination (MMSE) score of 12 to 24 (inclusive). 4. Is using acetylcholinesterase inhibitors (AChEI) therapy (e.g., donepezil, galantamine, or rivastigmine) for management of AD dementia. 5. Has a designated study partner/caregiver who can fulfill the requirements of this study.

4.2 Exclusion Criteria 1. Has a known history of stroke or cerebrovascular disease. 2. Has diagnosis of a clinically relevant central nervous system (CNS) disease other than AD dementia or has structural brain disease. 3. Has a history of seizures or epilepsy. 4. Has any other major CNS trauma, or infections that affect brain function (e.g., HIV, syphilis, and/or neurological sequelae of COVID-19).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-1167 oral capsules at 0.3 mg, 1 mg, or 3 mg, taken once daily (QD). **Route of Administration:** Oral. **Duration of Treatment:** Up to approximately 24 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background acetylcholinesterase inhibitors (AChEI) therapy. **Prohibited:** Medications or conditions that negatively impact cognition or cognitive status chronically.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, MMSE evaluation, Medical History, Verification of study partner
Baseline	Day 0	Randomization, First Dose, Baseline ADAS-Cog11 and ADCS-ADL Assessments
Interim 1	Week 12	Safety Labs, Efficacy Assessment (ADAS-Cog11, ADCS-ADL), Caregiver Survey
End of Study	Week 24	Final Efficacy Assessments (ADAS-Cog11, ADCS-ADL), Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in the Alzheimer's Disease Assessment Scale-11-item Cognitive Subscale (ADAS-Cog11) Total Score at Week 24. **Measurement Method:** Clinical assessment scale (evaluates memory, orientation, attention, reasoning, language, and constructional praxis) ranging from 0 to 70.

7.2 Secondary & Exploratory Endpoints * Change from baseline in the Alzheimer's Disease Cooperative Study - Activities of Daily Living (ADCS-ADL) score at Week 24. * Change from baseline in

ADAS-Cog11 score at Week 12. * Change from baseline in ADCS-ADL score at Week 12.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via study partner/caregiver interview, clinical evaluation, and physical exams. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee utilized to review accumulating safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome measurements; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Appropriately powered to detect statistically significant differences in the ADAS-Cog11 total score change from baseline between the MK-1167 dose groups and placebo. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) approach for continuous longitudinal endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to verify investigator compliance and data integrity. **Audit Trail:** Validated Electronic Data Capture (EDC) systems with full tracking, eCRF locking, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-1167-008 **Registry Number:** NCT06721156 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study to Evaluate the Efficacy and Safety of MK-1167 in Participants With Alzheimer's Disease Dementia **Official Regulatory Title:** A Phase 2 Randomized, Placebo-Controlled, Double-Blind, Parallel-group Study to Evaluate the Efficacy and Safety of MK-1167 as Adjunctive Therapy in Participants With Mild to Moderate Alzheimer's Disease Dementia

12. Clinical Framework (Alzheimer's Disease Specific)

Primary Condition: Alzheimer's Disease (AD) Dementia **Diagnosis Threshold:** MMSE score of 12 to 24 (Stage 4 or Stage 5 AD) **Medical Methodology:** Standard Acetylcholinesterase Inhibitor (AChEI)

Therapy **Optimization Target:** Improvement in cognition (ADAS-Cog11) and activities of daily living (ADCS-ADL)

13. Study Procedures & Methodology

Management Pathway: Adjunctive experimental therapy (MK-1167) added to standard AD dementia care **Education Protocol (HCP):** Administration guidelines for ADAS-Cog11 and ADCS-ADL scales **Education Protocol (Patient):** Comprehensive trial guidance utilizing the designated study partner/caregiver **Quality Audit Mechanism:** Caregiver reporting accuracy checks & study drug adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 24 weeks of active treatment **Onsite Visit Cadence:** Baseline, Week 12, and Week 24 **Remote Monitoring Frequency:** Standard protocol-defined safety follow-ups **Checklist Review Interval:** Every scheduled study visit via study partner input

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				